

Two-level evaluation and traceability workflow

Architecture path

AlexNet layer YAML

Workload dimensions



Timeloop

Mapping cycles utilization
and activity



Accelergy + CACTI

Component energies



Parser + validator

CSV results and traceability

Independent evidence paths; percentages are not combined

Circuit path

PTM + link netlist

Device and interconnect
parameters



ngspice 42

Transient waveforms and
measures



Parser + validator

Energy per toggle and delay



**Cross-level
interpretation**

Bounded architectural
conclusions

